Docker and Kubernetes: Habit Tracker

Group 8 Team

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Overview

The purpose of this project is to build a containerized web application using Docker and deploy it on a Kubernetes cluster hosted on the Chameleon Cloud. The application will serve as a Habit Tracker, allowing users to log daily habits and visualize their progress. This project demonstrates how cloud-native technologies like Docker and Kubernetes can support scalable, real-world applications.

Goals

1. **Understand Docker Containerization:** Learn to package a multi-component web application (frontend, backend, database) into isolated Docker containers.

2. **Master Kubernetes Orchestration:** Gain hands-on experience with deploying, managing, and scaling containerized applications on a Kubernetes cluster.

3. **Implement a Production-Ready Deployment:** Successfully set up a Kubernetes cluster on a cloud environment and deploy the habit tracker application with considerations for high availability and persistence.

Testbed and Software Tools

1. **Containerization:** Docker
2. **Orchestration:** Kubernetes
3. **Cloud Environment:** Chameleon

4. **Backend Development:** Python (Flask — lightweight and well-suited for small web services)
5. **Frontend Development:** HTML, CSS, and JavaScript
6. **Database:** PostgreSQL (for structured data with relational integrity)

Dataset

- **User-generated habit data:** Stored using the PostgreSQL Database
- **Dataset from our team**: We will make up some data so we can test our habit tracker effectively.